

Zstate Complete Framework - 4 Phases & 185 Tasks

Zstate Equity Research Agent Benchmark

Complete Framework - 4 Phases & 185 Task Definitions (v0.2)

Prepared for: Zstate.ai | Confidential

1. Executive Summary

Zstate builds agentic AI datasets - Task, Trajectory, and Reward loops - using credentialed domain experts. This document defines the complete equity research benchmark: four implementation phases, three-layer scoring, 185 micro-tasks from SEC/IR data through financial models to compliant investment output, and the 15-task MVD pilot scope.

2. Four-Phase Framework

Phase 1 - Foundation & Data Corpus

Timeline: Weeks 1-2

Map tickers to CIK; ingest SEC EDGAR (10-K, 10-Q, 8-K)

Ingest earnings transcripts (API primary, IR fallback)

Load FX reference rates (FRED); build section index

Expert spot-check 10% of transcripts; lock corpus manifest

Deliverable: Deliverable: corpus_pilot_manifest.json (~40-60 docs for 5 companies)

Phase 2 - Task Construction & Expert Validation

Timeline: Weeks 2-8

Author tasks from archetype templates (footnote, guidance, FX)

Build ground truth with 100% citation coverage

Define gold minimal section sets (not brittle tool sequences)

Write Layer 1 Python verification scripts

CFA peer review and publish to Task Registry

Deliverable: Deliverable: 15 MVD tasks (5 companies x 3 archetypes); scale to 185 over v0.1-v0.5

Phase 3 - Agent Evaluation & Trajectory Capture

Timeline: Weeks 3-10

Run model-agnostic agents via adapter layer

Execute task-type workflows (F/M/C - not universal 4-stage on all tasks)

Log full trajectories: thought, tool I/O, sections accessed

Detect fractures: LOOP_TOOL, SIGN_ERR, HALLUC_FILL, etc.

3 runs per task per model for pilot calibration

Deliverable: Deliverable: trajectory JSONL per run; fracture report

Phase 4 - Scoring, Release & Trajectory Dataset

Timeline: Weeks 8-10+

Layer 1: programmatic numeric verification

Layer 2: automated section recall + expert calibration sample

Layer 3: citation audit; FINRA/mandates on Type C tasks only

Publish benchmark_v0.1 with leaderboard

Export curated trajectories for Phase 2 training product

Deliverable: Deliverable: reward vectors, leaderboard, trajectory dataset for SFT/RL

3. Task-Type Taxonomy (Agent Workflow)

4. Three-Layer Reward Model

5. Gold Trajectories & Anti-Patterns

Gold trajectories use a minimal section set (not brittle tool sequences). The gold trajectory acts as a multi-objective optimization signal for reinforcement learning (RL), specifically rewarding path efficiency and section recall accuracy.

Anti-patterns (required in every gold path JSON): Expert-defined negative signals for the reward model - e.g., load_entire_10k (path bloat), skip_accounting_policies_note (section recall failure), produce_buy_hold_sell_recommendation on Type F tasks.

6. MVD Pilot - 15 Tasks

7. Full Task Catalog - 185 Tasks (Detailed)

Each task below includes ID, name, description, inputs, outputs, dependencies, pass/fail criteria, and difficulty. Organized by layer.

Layer 0 - Data Acquisition

D-001: Map Ticker to CIK

Description: Resolve public ticker symbol to SEC Central Index Key and legal entity name. |

Inputs: Ticker (e.g., GOOGL) |

Outputs: `{ cik, entity\_name, exchange, ticker }` |

Dependencies: None |

Pass criteria: CIK matches SEC EDGAR master |

Fail conditions: Wrong entity (e.g., GOOG vs GOOGL class confusion without disclosure) |

Difficulty: L1 |

D-002: Fetch Latest 10-K for Fiscal Year

Description: Download and index the authoritative 10-K for a specified fiscal year, preferring latest amendment. |

Inputs: Ticker, fiscal year (e.g., FY2024) |

Outputs: Raw filing, `doc\_id`, filing date, checksum |

Dependencies: D-001 |

Pass criteria: Correct FY; 10-K/A supersedes original if amended |

Fail conditions: Wrong year; pre-amendment version used |

Difficulty: L1 |

D-003: Fetch Trailing Eight 10-Q Filings

Description: Retrieve eight most recent quarterly filings for trend and guidance-drift analysis. |

Inputs: Ticker |

Outputs: Indexed list of 8 `doc\_id`s with fiscal quarters |

Dependencies: D-001, D-005 |

Pass criteria: No missing quarters in window |

Fail conditions: Gap quarter; wrong fiscal calendar |

Difficulty: L1 |

D-004: Detect Filing Amendments

Description: Identify 10-K/A, 10-Q/A and mark superseded documents. |

Inputs: CIK, document list |

Outputs: Amendment graph: `{ original\_doc\_id, superseded\_by }` |

Dependencies: D-002, D-003 |

Pass criteria: Latest amendment flagged as canonical |

Fail conditions: Original used when amendment exists |

Difficulty: L2 |

D-005: Parse Fiscal Calendar

Description: Determine fiscal year-end month (Dec, Jun, Sep, etc.) from 10-K cover page. |

Inputs: 10-K |

Outputs: `{ fy\_end\_month, period\_convention }` |

Dependencies: D-002 |

Pass criteria: FY end matches SEC header |

Fail conditions: Calendar assumption causes quarter mis-map |

Difficulty: L1 |

D-006: Extract Income Statement (3-5 Years)

Description: Extract multi-year IS: revenue through net income, EPS if reported. |

Inputs: 10-K series |

Outputs: Structured IS JSON with periods and units |

Dependencies: D-002, D-009 |

Pass criteria: Line items match filing; units correct |

Fail conditions: Wrong period; thousands vs millions error |

Difficulty: L2 |

D-007: Extract Balance Sheet (2-5 Years)

Description: Extract assets, liabilities, equity with current/non-current split. |

Inputs: 10-K, latest 10-Q |

Outputs: Structured BS JSON |

Dependencies: D-002, D-003, D-009 |

Pass criteria: $A = L + E$ on extracted numbers |

Fail conditions: Missing lease lines; misclassified current debt |

Difficulty: L2 |

D-008: Extract Cash Flow Statement (3-5 Years)

Description: Extract OCF, investing, financing sections including CapEx, buybacks, dividends. |

Inputs: 10-K series |

Outputs: Structured CF JSON |

Dependencies: D-002, D-009 |

Pass criteria: CapEx and SBC in CF reconcile to footnotes +/-materiality |

Fail conditions: Sign inversion on CapEx or buybacks |

Difficulty: L2 |

D-009: Identify Unit Convention

Description: Detect whether tables are in thousands, millions, or billions. |

Inputs: Any filing financial table |

Outputs: `{ unit\_multiplier, unit\_label }` |

Dependencies: None |

Pass criteria: Multiplier matches table header |

Fail conditions: 1000x magnitude error downstream |

Difficulty: L1 |

D-010: Distinguish GAAP vs Non-GAAP Labels

Description: Parse IR earnings deck/release and map non-GAAP metrics to GAAP equivalents. |

Inputs: IR PDF, 10-Q |

Outputs: Label mapping table |

Dependencies: D-013 |

Pass criteria: No conflation of adjusted vs GAAP in storage |

Fail conditions: Stores "Adjusted EPS" as GAAP EPS |

Difficulty: L2 |

D-011: Pull Earnings Transcript

Description: Ingest earnings call transcript for specified fiscal quarter (API primary, IR fallback). |

Inputs: Ticker, fiscal quarter |

Outputs: Transcript text, `doc\_id`, event date, speaker index |

Dependencies: D-001, D-005 |

Pass criteria: Correct quarter and call date |

Fail conditions: Wrong quarter; wrong company |

Difficulty: L1 |

D-012: Align Transcript Quarter to 10-Q

Description: Link transcript fiscal period to corresponding 10-Q `doc\_id`. |

Inputs: Transcript metadata, 10-Q list |

Outputs: `{ transcript\_doc\_id, ten\_q\_doc\_id, fiscal\_period }` |

Dependencies: D-003, D-011 |

Pass criteria: Same fiscal period on both |

Fail conditions: Off-by-one quarter |

Difficulty: L2 |

D-013: Fetch Investor Presentation

Description: Download IR earnings presentation matching earnings date. |

Inputs: Ticker, earnings date |

Outputs: Deck PDF, `doc\_id` |

Dependencies: D-011 |

Pass criteria: Deck date matches earnings |

Fail conditions: Wrong event deck |

Difficulty: L1 |

D-014: Extract Supplemental KPI Table

Description: Extract KPIs from IR supplemental (subs, units, ARPU, etc.). |

Inputs: IR deck/Excel, 10-Q |

Outputs: KPI time series JSON |

Dependencies: D-013, D-003 |

Pass criteria: KPIs cross-check to 10-Q where disclosed |

Fail conditions: Invented KPI not in source |

Difficulty: L2 |

D-015: Build Peer Universe

Description: Select 5-8 comparable companies with written GICS/segment rationale. |

Inputs: Target ticker, sector |

Outputs: Peer list + rationale memo |

Dependencies: D-001 |

Pass criteria: Peers same industry; rationale cites business model |

Fail conditions: Cross-sector peer without justification |

Difficulty: L2 |

D-016: Pull Peer 10-K Filings

Description: Ingest latest 10-K for each peer in universe. |

Inputs: Peer list |

Outputs: Peer filing bundle |

Dependencies: D-015, D-002 |

Pass criteria: All peers have current FY filing |

Fail conditions: Missing peer data |

Difficulty: L1 |

D-017: Load FX Average Rates

Description: Build weighted-average FX table for currencies disclosed in geographic segments. |

Inputs: 10-K FX note, FRED rates |

Outputs: `{ pair, period, avg\_rate, rate\_type }` table |

Dependencies: D-002 |

Pass criteria: Uses weighted avg when footnote specifies; flags spot misuse |

Fail conditions: Spot rate used for organic growth calc |

Difficulty: L2 |

D-018: Load Shares Outstanding

Description: Extract basic and diluted share count by period. |

Inputs: 10-K, 10-Q, EPS footnote |

Outputs: Share count series |

Dependencies: D-002, D-003 |

Pass criteria: Diluted used for EPS; period-end for market cap noted |

Fail conditions: Wrong share class; wrong period |

Difficulty: L2 |

D-019: Load Net Debt Components

Description: Compute net debt: ST debt + LT debt + leases - cash & equivalents. |

Inputs: BS, debt footnote |

Outputs: `{ net\_debt, components[] }` |

Dependencies: D-007 |

Pass criteria: Matches footnote reconciliation |

Fail conditions: Omits ST debt or lease liabilities |

Difficulty: L2 |

D-020: Flag Restatement / Accounting Change

Description: Scan footnotes for restatements, accounting policy changes, prior period adjustments. |

Inputs: 10-K notes |

Outputs: Change log with impacted line items |

Dependencies: D-002 |

Pass criteria: All material changes logged |

Fail conditions: Misses Note 1 policy change |

Difficulty: L2 |

D-021: Ingest 8-K Material Event

Description: Fetch and summarize 8-K filings in date window (M&A, impairment, leadership). |

Inputs: Ticker, date range |

Outputs: Event summary list |

Dependencies: D-001 |

Pass criteria: Material events captured with date and item code |

Fail conditions: Misses EX-99.1 earnings 8-K |

Difficulty: L2 |

D-022: Cross-Check 8-K Press Release vs 10-Q

Description: Compare EX-99.1 earnings release numbers to subsequently filed 10-Q. |

Inputs: 8-K exhibit, 10-Q |

Outputs: Discrepancy report |

Dependencies: D-021, D-003 |

Pass criteria: Material diffs explained (reclassification, subsequent events) |

Fail conditions: Ignores pre-release vs filed gap |

Difficulty: L3 |

D-023: Version Corpus Snapshot

Description: Lock all documents for benchmark run with root checksum manifest. |

Inputs: Full document set |

Outputs: `corpus\_v1` manifest |

Dependencies: D-002 through D-022 |

Pass criteria: Immutable manifest; all doc checksums |

Fail conditions: Drift between eval runs |

Difficulty: L1 |

D-024: Cite Every Extracted Number

Description: Attach `{ doc\_id, page, snippet, table\_id }` to every extracted data point. |

Inputs: Any extraction task output |

Outputs: Citation-enriched data package |

Dependencies: Parallel to D-006-D-019 |

Pass criteria: 100% citation coverage |

Fail conditions: Any number without auditable path |

Difficulty: L2 |

Layer 1 - Financial Statement Intelligence

FS-001: Identify Non-GAAP Reconciliations

Description: Locate all non-GAAP metrics and reconciliation tables in earnings release and 10-Q. |

Inputs: IR release, 10-Q |

Outputs: List of non-GAAP metrics + reconciliation refs |

Dependencies: D-010, D-003 |

Pass criteria: All adjusted metrics mapped to GAAP |

Fail conditions: Uses adjusted without reconciliation |

Difficulty: L2 | ****Reward**** | L2 |

FS-002: Rebuild GAAP -> Adjusted EBITDA Bridge

Description: Construct full bridge from GAAP net income to company-reported adjusted EBITDA. |

Inputs: Reconciliation tables |

Outputs: Line-by-line bridge with citations |

Dependencies: FS-001 |

Pass criteria: Bridge sums correctly |

Fail conditions: Omits stock comp or restructuring add-back |

Difficulty: L2 | ****Reward**** | L1, L2 |

FS-003: Separate Recurring vs One-Time Operating Items

Description: Classify operating income components as recurring or one-time. |

Inputs: IS, footnotes, 8-K |

Outputs: Core operating income vs adjustments |

Dependencies: FS-002, D-020 |

Pass criteria: One-times identified with footnote source |

Fail conditions: Treats asset sale gain as recurring |

Difficulty: L2 | ****Reward**** | L2 |

FS-004: Footnote Reconciliation - Segment vs Consolidated

Description: Reconcile segment revenue/OP to consolidated; resolve eliminations and reclassifications in footnotes. |

Inputs: Segment table, accounting policy note |

Outputs: Reconciliation table + discrepancy flag |

Dependencies: D-006, D-024 |

Pass criteria: Segments tie to consolidated +/- disclosed eliminations |

Fail conditions: Misses footnote-only reclassification (****MVD archetype****) |

Difficulty: L3 | ****Reward**** | L1, L2 |

FS-005: Footnote Reconciliation - Inventory Costing Change

Description: Quantify impact of inventory accounting change on COGS and margins. |

Inputs: Note 1/2, IS |

Outputs: Impact quantification by period |

Dependencies: D-020, D-006 |

Pass criteria: Impact matches footnote disclosure |

Fail conditions: Ignores retrospective application |

Difficulty: L3 | ****Reward**** | L1, L2 |

FS-006: Extract Stock-Based Compensation

Description: Pull SBC from footnote and CF statement; note cash tax benefit if any. |

Inputs: 10-K notes, CF |

Outputs: SBC by year; cash vs non-cash |

Dependencies: D-006, D-008 |

Pass criteria: Matches footnote total |

Fail conditions: Confuses SBC with other equity comp |

Difficulty: L2 | ****Reward**** | L1 | ****Sectors**** | Tech |

FS-007: Allocate SBC to Segments

Description: Allocate enterprise SBC to segments when not directly reported. |

Inputs: SBC total, segment headcount/revenue proxy, footnotes |

Outputs: Segment SBC allocation with method |
Dependencies: FS-006 |
Pass criteria: Method disclosed; sums to total |
Fail conditions: Arbitrary split without rationale |
Difficulty: L3 | **Reward** | L2 | **Sectors** | Tech |

FS-008: Identify Capitalized Software / R&D

Description: Quantify capitalized development costs and amortization from footnotes. |
Inputs: Intangible assets note, R&D footnote |
Outputs: Capitalized vs expensed split |
Dependencies: D-008, footnotes |
Pass criteria: Ties to CF and BS intangible roll-forward |
Fail conditions: Misses cloud software capitalization |
Difficulty: L3 | **Reward** | L2 | **Sectors** | Tech (GOOGL, MSFT) |

FS-009: Content Amortization vs Cash Content Spend

Description: Compare content amortization (P&L/CF) to cash content investments on balance sheet. |
Inputs: NFLX/DIS content footnote, CF |
Outputs: Amort vs cash spend bridge |
Dependencies: D-008, content asset note |
Pass criteria: Distinguishes amort from cash spend |
Fail conditions: Uses amort as proxy for cash investment |
Difficulty: L3 | **Reward** | L2 | **Sectors** | Media |

FS-010: Reconcile Franchise vs Company-Operated Revenue

Description: Separate franchise royalties from company-operated store revenue and margins. |
Inputs: Segment/franchise disclosure |
Outputs: Split revenue and OP metrics |
Dependencies: D-006, franchise note |
Pass criteria: Matches 10-K franchise disclosure |
Fail conditions: Blends unlike margin profiles |
Difficulty: L2 | **Reward** | L2 | **Sectors** | Consumer (MCD, SBUX) |

FS-011: Lease-Adjusted Metrics

Description: Compute EBITDAR or adjust for ASC 842 lease accounting when comparing peers. |
Inputs: Lease footnote, IS |
Outputs: Lease-adjusted EBITDA metrics |
Dependencies: D-007, lease note |
Pass criteria: Lease liability/op lease cost handled per ASC 842 |
Fail conditions: Double-counts rent |
Difficulty: L3 | **Reward** | L2 | **Sectors** | Retail, restaurants |

FS-012: Organic Constant-Currency Revenue Growth

Description: Calculate organic CC revenue growth by geography using weighted-average FX. |
Inputs: Geographic revenue, FX table (D-017) |
Outputs: Organic CC growth by region and consolidated |
Dependencies: D-017, segment footnote |
Pass criteria: Uses weighted avg; separates M&A if disclosed |
Fail conditions: Spot rate; ignores acquisition impact (**MVD archetype**) |
Difficulty: L3 | **Reward** | L1, L2 |

FS-013: Price vs Volume vs Mix Decomposition

Description: Decompose revenue growth into price, volume, mix when company discloses. |
Inputs: MD&A, IR deck, 10-Q |
Outputs: P/V/M table |
Dependencies: D-006, D-013 |
Pass criteria: Components sum to total growth |
Fail conditions: Invents decomposition not disclosed |
Difficulty: L2 | **Reward** | L2 | **Sectors** | Consumer |

FS-014: Goodwill Impairment Triggers

Description: Identify goodwill/intangible impairment indicators and recorded charges. |

Inputs: Goodwill footnote, 8-K |

Outputs: Impairment history + trigger assessment |

Dependencies: D-007, D-021 |

Pass criteria: Impairments tied to segment reporting |

Fail conditions: Misses WBD-style legacy goodwill risk |

Difficulty: L2 | ****Reward**** | L2 | ****Sectors**** | Media |

FS-015: Debt Maturity Schedule

Description: Extract contractual debt maturity ladder from footnote. |

Inputs: Debt note |

Outputs: Maturity schedule by year |

Dependencies: D-007 |

Pass criteria: Matches total debt on BS |

Fail conditions: Omits revolving credit availability |

Difficulty: L2 | ****Reward**** | L1, L2 |

FS-016: Off-Balance-Sheet Commitments

Description: Quantify operating lease commitments, purchase obligations, guarantees. |

Inputs: Commitments & contingencies note |

Outputs: Commitment summary table |

Dependencies: D-002 |

Pass criteria: All material commitments captured |

Fail conditions: Ignores purchase obligations |

Difficulty: L2 | ****Reward**** | L2, L3 |

FS-017: Related-Party Transactions

Description: Identify related-party transactions from proxy and footnotes. |

Inputs: DEF 14A, footnotes |

Outputs: RPT list with amounts |

Dependencies: D-002 |

Pass criteria: Material RPTs disclosed |

Fail conditions: Misses proxy-only disclosure |

Difficulty: L2 | ****Reward**** | L3 |

FS-018: Verify Cash Flow Sign Conventions

Description: Confirm CapEx, buybacks, dividends, debt issuance shown with correct sign in CF and models. |

Inputs: CF statement, model CF |

Outputs: Sign validation report |

Dependencies: D-008 |

Pass criteria: CapEx negative in FCF calc; buybacks negative in financing |

Fail conditions: Sign inversion inflates FCF (****critical****) |

Difficulty: L2 | ****Reward**** | L1 (critical veto) |

FS-019: Reconcile Net Income -> OCF -> FCF

Description: Build and verify indirect method bridge from NI to OCF to unlevered FCF. |

Inputs: IS, CF, WC changes |

Outputs: Integrated bridge |

Dependencies: D-006, D-008, FS-018 |

Pass criteria: Bridge ties within +/-0.01% |

Fail conditions: WC change wrong sign |

Difficulty: L3 | ****Reward**** | L1 |

FS-020: Auditor Opinion & Going Concern

Description: Flag changes in auditor, qualified opinion, going concern paragraph. |

Inputs: 10-K audit report |

Outputs: Audit risk flags |

Dependencies: D-002 |

Pass criteria: Correct opinion type extracted |

Fail conditions: Misses going concern |

Difficulty: L1 | ****Reward**** | L3 |

FS-021: MD&A Narrative vs Quantitative Trend

Description: Compare management MD&A claims to reported numbers (guidance drift precursor). |

Inputs: MD&A, IS/CF trends |

Outputs: Narrative vs data alignment report |

Dependencies: D-006, D-008 |

Pass criteria: Flags contradictions |

Fail conditions: Accepts narrative without verification |

Difficulty: L2 | ****Reward**** | L2 |

FS-022: Change in Accounting Principle

Description: Identify and summarize new accounting standards adopted (Note 1). |

Inputs: Note 1 |

Outputs: Policy change memo |

Dependencies: D-020 |

Pass criteria: Effective date and impact stated |

Fail conditions: Misses retrospective restatement |

Difficulty: L2 | ****Reward**** | L2 |

FS-023: Segment OP vs Consolidated OP Reconciliation

Description: Reconcile sum of segment operating profit to consolidated OP including unallocated corporate. |

Inputs: Segment note |

Outputs: Reconciliation with corporate overhead |

Dependencies: FS-004 |

Pass criteria: Corporate overhead identified |

Fail conditions: Double-counts eliminations |

Difficulty: L3 | ****Reward**** | L1, L2 | ****Sectors**** | Conglomerates |

FS-024: Calculate NOPAT for ROIC

Description: Compute NOPAT and invested capital for ROIC analysis. |

Inputs: Normalized EBIT, tax rate, BS |

Outputs: `{ nopat, invested\_capital, roic }` |

Dependencies: FS-003, D-007 |

Pass criteria: Tax rate normalized; IC excludes excess cash per policy |

Fail conditions: Uses headline tax rate with one-time benefit |

Difficulty: L3 | ****Reward**** | L1, L2 |

Layer 2A - Three-Statement Model

M3-001: Build 5-Year Historical Income Statement

Description: Populate historical IS from extracted filings. | ****Dependencies**** | D-006, FS-003 | ****Pass**** | Matches filings cited | ****Fail**** | Mixed annual/quarterly | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: D-006, FS-003 | ****Pass**** | Matches filings cited | ****Fail**** | Mixed annual/quarterly | ****Difficulty**** | L2 |

Pass criteria: Matches filings cited | ****Fail**** | Mixed annual/quarterly | ****Difficulty**** | L2 |

Fail conditions: Mixed annual/quarterly | ****Difficulty**** | L2 |

Difficulty: L2 |

M3-002: Build 5-Year Historical BS and CF

Description: Populate historical balance sheet and cash flow. | ****Dependencies**** | D-007, D-008 | ****Pass**** | A=L+E; CF ties | ****Fail**** | Broken links | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: D-007, D-008 | ****Pass**** | A=L+E; CF ties | ****Fail**** | Broken links | ****Difficulty**** | L2 |

Pass criteria: A=L+E; CF ties | ****Fail**** | Broken links | ****Difficulty**** | L2 |

Fail conditions: Broken links | ****Difficulty**** | L2 |

Difficulty: L2 |

M3-003: Identify Revenue Drivers by Segment

Description: Map revenue to segment, geographic, or unitxprice drivers. | ****Dependencies**** | M3-001, segment note | ****Pass**** | Driver map cited | ****Fail**** | Single growth rate with no segment detail | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: M3-001, segment note | ****Pass**** | Driver map cited | ****Fail**** | Single growth rate with no segment detail | ****Difficulty**** | L2 |

Pass criteria: Driver map cited | ****Fail**** | Single growth rate with no segment detail | ****Difficulty**** | L2 |

Fail conditions: Single growth rate with no segment detail | ****Difficulty**** | L2 |

Difficulty: L2 |

M3-004: Forecast Revenue 5 Years (Segment-Level)

Description: Project revenue by segment/geography with stated assumptions. | ****Dependencies**** | M3-003, FS-012 | ****Pass**** | Segments sum to total | ****Fail**** | Top-down only with no segment build | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-003, FS-012 | ****Pass**** | Segments sum to total | ****Fail**** | Top-down only with no segment build | ****Difficulty**** | L3 |

Pass criteria: Segments sum to total | ****Fail**** | Top-down only with no segment build | ****Difficulty**** | L3 |

Fail conditions: Top-down only with no segment build | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-005: Forecast Gross Margin (Commodity Sensitivity)

Description: Model gross margin with input cost drivers where relevant. | ****Dependencies**** | M3-004 | ****Pass**** | Sensitivity to commodity shown | ****Fail**** | Flat margin without rationale | ****Difficulty**** | L3 | ****Sectors**** | PEP, KO, MDLZ |

Inputs: -

Outputs: -

Dependencies: M3-004 | ****Pass**** | Sensitivity to commodity shown | ****Fail**** | Flat margin without rationale | ****Difficulty**** | L3 | ****Sectors**** | PEP, KO, MDLZ |

Pass criteria: Sensitivity to commodity shown | ****Fail**** | Flat margin without rationale | ****Difficulty**** | L3 | ****Sectors**** | PEP, KO, MDLZ |

Fail conditions: Flat margin without rationale | ****Difficulty**** | L3 | ****Sectors**** | PEP, KO, MDLZ |

Difficulty: L3 | ****Sectors**** | PEP, KO, MDLZ |

M3-006: Forecast OpEx with Operating Leverage

Description: Project R&D, S&M, G&A as % revenue or absolute with leverage thesis. | ****Dependencies**** | M3-004 | ****Pass**** | OpEx ties to historical ratios bounded | ****Fail**** | OpEx declines forever without limit | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-004 | ****Pass**** | OpEx ties to historical ratios bounded | ****Fail**** | OpEx declines forever without limit | ****Difficulty**** | L3 |

Pass criteria: OpEx ties to historical ratios bounded | ****Fail**** | OpEx declines forever without limit | ****Difficulty**** | L3 |

Fail conditions: OpEx declines forever without limit | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-007: Build D&A from PP&E Roll-Forward

Description: Link CapEx and D&A through PP&E schedule. | ****Dependencies**** | M3-002, M3-008 | ****Pass**** | PP&E roll ties | ****Fail**** | D&A disconnected from assets | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-002, M3-008 | ****Pass**** | PP&E roll ties | ****Fail**** | D&A disconnected from assets | ****Difficulty**** | L3 |

Pass criteria: PP&E roll ties | ****Fail**** | D&A disconnected from assets | ****Difficulty**** | L3 |

Fail conditions: D&A disconnected from assets | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-008: Build CapEx Schedule (Maintenance + Growth)

Description: Split CapEx into maintenance and growth; align with mgmt guidance. | ****Dependencies**** | M3-004, D-008 | ****Pass**** | CapEx/Revenue ratio bounded vs history | ****Fail**** | Ignores AI/cloud capex guidance (tech) | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-004, D-008 | ****Pass**** | CapEx/Revenue ratio bounded vs history | ****Fail**** | Ignores AI/cloud capex guidance (tech) | ****Difficulty**** | L3 |

Pass criteria: CapEx/Revenue ratio bounded vs history | ****Fail**** | Ignores AI/cloud capex guidance (tech) | ****Difficulty**** | L3 |

Fail conditions: Ignores AI/cloud capex guidance (tech) | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-009: Build NWC Schedule (DSO, DIO, DPO)

Description: Forecast working capital from turnover ratios. | ****Dependencies**** | M3-002 | ****Pass**** | ?NWC feeds CF correctly | ****Fail**** | NWC change wrong sign in FCF | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-002 | ****Pass**** | ?NWC feeds CF correctly | ****Fail**** | NWC change wrong sign in FCF | ****Difficulty**** | L3 |

Pass criteria: ?NWC feeds CF correctly | ****Fail**** | NWC change wrong sign in FCF | ****Difficulty**** | L3 |

Fail conditions: NWC change wrong sign in FCF | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-010: PP&E Roll-Forward Link

Description: Verify Beginning + CapEx - D&A = Ending PP&E each year. | ****Dependencies**** | M3-007, M3-008 | ****Pass**** | Roll-forward exact | ****Fail**** | Broken link | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: M3-007, M3-008 | ****Pass**** | Roll-forward exact | ****Fail**** | Broken link | ****Difficulty**** | L2 |

Pass criteria: Roll-forward exact | ****Fail**** | Broken link | ****Difficulty**** | L2 |

Fail conditions: Broken link | ****Difficulty**** | L2 |

Difficulty: L2 |

M3-011: Debt Schedule with Interest Expense

Description: Model debt balances, interest at effective rate, mandatory amortization. | ****Dependencies**** | FS-015, D-019 | ****Pass**** | Interest ties to footnote rate +/-10bps | ****Fail**** | Interest on wrong base | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: FS-015, D-019 | ****Pass**** | Interest ties to footnote rate +/-10bps | ****Fail**** | Interest on wrong base | ****Difficulty**** | L3 |

Pass criteria: Interest ties to footnote rate +/-10bps | ****Fail**** | Interest on wrong base | ****Difficulty**** | L3 |

Fail conditions: Interest on wrong base | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-012: Model SBC and Share Count Dilution

Description: Forecast SBC and diluted share count path. | ****Dependencies**** | FS-006, D-018 | ****Pass**** | Dilution matches repurchase offset | ****Fail**** | Flat shares ignoring SBC | ****Difficulty**** | L3 | ****Sectors**** | Tech |

Inputs: -

Outputs: -

Dependencies: FS-006, D-018 | ****Pass**** | Dilution matches repurchase offset | ****Fail**** | Flat shares ignoring SBC | ****Difficulty**** | L3 | ****Sectors**** | Tech |

Pass criteria: Dilution matches repurchase offset | ****Fail**** | Flat shares ignoring SBC | ****Difficulty**** | L3 | ****Sectors**** | Tech |

Fail conditions: Flat shares ignoring SBC | ****Difficulty**** | L3 | ****Sectors**** | Tech |

Difficulty: L3 | ****Sectors**** | Tech |

M3-013: Model Share Repurchase Program

Description: Apply authorization-based buyback to equity and share count. | ****Dependencies**** | M3-012, CF | ****Pass**** | Buybacks tie to CF financing | ****Fail**** | Buybacks exceed authorization without flag | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: M3-012, CF | ****Pass**** | Buybacks tie to CF financing | ****Fail**** | Buybacks exceed authorization without flag | ****Difficulty**** | L2 |

Pass criteria: Buybacks tie to CF financing | ****Fail**** | Buybacks exceed authorization without flag | ****Difficulty**** | L2 |

Fail conditions: Buybacks exceed authorization without flag | ****Difficulty**** | L2 |

Difficulty: L2 |

M3-014: Model Dividend Payout Schedule

Description: Project DPS and total dividends from payout policy. | ****Dependencies**** | M3-015, CF | ****Pass**** | Dividends <= FCF or flagged | ****Fail**** | Unsustainable payout ignored | ****Difficulty**** | L2 | ****Sectors**** | PEP, KO, MCD |

Inputs: -

Outputs: -

Dependencies: M3-015, CF | ****Pass**** | Dividends <= FCF or flagged | ****Fail**** | Unsustainable payout ignored | ****Difficulty**** | L2 | ****Sectors**** | PEP, KO, MCD |

Pass criteria: Dividends <= FCF or flagged | ****Fail**** | Unsustainable payout ignored | ****Difficulty**** | L2 | ****Sectors**** | PEP, KO, MCD |

Fail conditions: Unsustainable payout ignored | ****Difficulty**** | L2 | ****Sectors**** | PEP, KO, MCD |

Difficulty: L2 | ****Sectors**** | PEP, KO, MCD |

M3-015: Complete Projected Cash Flow (Indirect Method)

Description: Build full projected CF from NI through OCF, ICF, FCF. | ****Dependencies**** | M3-001-M3-014 | ****Pass**** | Ending cash = BS cash | ****Fail**** | CF not linked to BS | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-001-M3-014 | ****Pass**** | Ending cash = BS cash | ****Fail**** | CF not linked to BS | ****Difficulty**** | L3 |

Pass criteria: Ending cash = BS cash | ****Fail**** | CF not linked to BS | ****Difficulty**** | L3 |

Fail conditions: CF not linked to BS | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-016: Verify A = L + E All Periods

Description: Balance sheet identity check every historical and forecast year. | ****Dependencies**** | M3-015 | ****Pass**** | Zero balance sheet error all periods | ****Fail**** | Any period fails (****critical****) | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: M3-015 | ****Pass**** | Zero balance sheet error all periods | ****Fail**** | Any period fails (****critical****) | ****Difficulty**** | L2 |

Pass criteria: Zero balance sheet error all periods | ****Fail**** | Any period fails (****critical****) | ****Difficulty**** | L2 |

Fail conditions: Any period fails (****critical****) | ****Difficulty**** | L2 |

Difficulty: L2 |

M3-017: Calculate Unlevered Free Cash Flow

Description: Derive UFCF = EBIT(1-t) + D&A - CapEx - ?NWC from model. | ****Dependencies**** | M3-015, M3-016 | ****Pass**** | UFCF matches manual Python verify +/-0.01% | ****Fail**** | Uses net income shortcut | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-015, M3-016 | ****Pass**** | UFCF matches manual Python verify +/-0.01% | ****Fail**** | Uses net income shortcut | ****Difficulty**** | L3 |

Pass criteria: UFCF matches manual Python verify +/-0.01% | ****Fail**** | Uses net income shortcut | ****Difficulty**** | L3 |

Fail conditions: Uses net income shortcut | ****Difficulty**** | L3 |

Difficulty: L3 |

M3-018: Scenario - Recession Case (-10% Revenue)

Description: Downside IS/CF with recession assumptions. | ****Dependencies**** | M3-004-M3-017 | ****Pass**** | Scenario documented and still balances | ****Fail**** | Breaks balance sheet | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-004-M3-017 | **Pass** | Scenario documented and still balances | **Fail** | Breaks balance sheet | **Difficulty** | L3 |

Pass criteria: Scenario documented and still balances | **Fail** | Breaks balance sheet | **Difficulty** | L3 |

Fail conditions: Breaks balance sheet | **Difficulty** | L3 |

Difficulty: L3 |

M3-019: Scenario - Margin Expansion Upside

Description: Upside case with explicit margin drivers. | **Dependencies** | M3-004-M3-017 | **Pass** | Bounded vs peer margins | **Fail** | Unrealistic margin target | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-004-M3-017 | **Pass** | Bounded vs peer margins | **Fail** | Unrealistic margin target | **Difficulty** | L3 |

Pass criteria: Bounded vs peer margins | **Fail** | Unrealistic margin target | **Difficulty** | L3 |

Fail conditions: Unrealistic margin target | **Difficulty** | L3 |

Difficulty: L3 |

M3-020: Export Model Assumptions with Citations

Description: Produce assumption log: every input linked to source. | **Dependencies** | M3-001-M3-019 | **Pass** | 100% material assumptions cited | **Fail** | Uncited key driver | **Difficulty** | L2 | **Reward** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-001-M3-019 | **Pass** | 100% material assumptions cited | **Fail** | Uncited key driver | **Difficulty** | L2 | **Reward** | L3 |

Pass criteria: 100% material assumptions cited | **Fail** | Uncited key driver | **Difficulty** | L2 | **Reward** | L3 |

Fail conditions: Uncited key driver | **Difficulty** | L2 | **Reward** | L3 |

Difficulty: L2 | **Reward** | L3 |

Layer 2B - DCF Valuation

DCF-001: Calculate CAPM Cost of Equity

Description: $K_e = R_f + \beta \times ERP$. | **Inputs** | FRED R_f , beta, ERP assumption | **Dependencies** | D-018, market data | **Pass** | Formula correct; inputs cited | **Fail** | Hardcoded K_e | **Difficulty** | L2 |

Inputs: FRED R_f , beta, ERP assumption | **Dependencies** | D-018, market data | **Pass** | Formula correct; inputs cited | **Fail** | Hardcoded K_e | **Difficulty** | L2 |

Outputs: -

Dependencies: D-018, market data | **Pass** | Formula correct; inputs cited | **Fail** | Hardcoded K_e | **Difficulty** | L2 |

Pass criteria: Formula correct; inputs cited | **Fail** | Hardcoded K_e | **Difficulty** | L2 |

Fail conditions: Hardcoded K_e | **Difficulty** | L2 |

Difficulty: L2 |

DCF-002: Justify Beta Selection

Description: Document 2-yr vs 5-yr regression choice vs peer beta. | **Dependencies** | DCF-001 | **Pass** | Written rationale | **Fail** | Beta without source | **Difficulty** | L2 | **Reward** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-001 | **Pass** | Written rationale | **Fail** | Beta without source | **Difficulty** | L2 | **Reward** | L2 |

Pass criteria: Written rationale | **Fail** | Beta without source | **Difficulty** | L2 | **Reward** | L2 |

Fail conditions: Beta without source | **Difficulty** | L2 | **Reward** | L2 |

Difficulty: L2 | **Reward** | L2 |

DCF-003: Calculate Pre-Tax Cost of Debt

Description: K_d from interest expense / average debt or footnote yield. | **Dependencies** | FS-015, M3-011 | **Pass** | Matches footnote +/-25bps | **Fail** | Uses coupon on wrong debt balance | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: FS-015, M3-011 | **Pass** | Matches footnote +/-25bps | **Fail** | Uses coupon on wrong debt balance | **Difficulty** | L2 |

Pass criteria: Matches footnote +/-25bps | **Fail** | Uses coupon on wrong debt balance | **Difficulty** | L2 |

Fail conditions: Uses coupon on wrong debt balance | **Difficulty** | L2 |

Difficulty: L2 |

DCF-004: Compute WACC

Description: Weight K_e and K_d by market values; tax-adjust K_d . | **Dependencies** | DCF-001, DCF-003, D-019 | **Pass** | WACC within peer range | **Fail** | Book value weights used | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: DCF-001, DCF-003, D-019 | **Pass** | WACC within peer range | **Fail** | Book value weights used | **Difficulty** | L3 |

Pass criteria: WACC within peer range | **Fail** | Book value weights used | **Difficulty** | L3 |

Fail conditions: Book value weights used | **Difficulty** | L3 |

Difficulty: L3 |

DCF-005: Project UFCF 5-10 Years

Description: Pull UFCF from 3-statement model. | **Dependencies** | M3-017 | **Pass** | Matches M3-017 exactly | **Fail** | Skips model; guesses FCF | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: M3-017 | **Pass** | Matches M3-017 exactly | **Fail** | Skips model; guesses FCF | **Difficulty** | L3 |

Pass criteria: Matches M3-017 exactly | **Fail** | Skips model; guesses FCF | **Difficulty** | L3 |

Fail conditions: Skips model; guesses FCF | **Difficulty** | L3 |

Difficulty: L3 |

DCF-006: Select Terminal Value Method

Description: Choose Gordon growth vs exit multiple with written rationale. | **Dependencies** | DCF-005 | **Pass** | Method matches business maturity | **Fail** | Gordon on high-growth tech without bounds | **Difficulty** | L2 | **Reward** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-005 | **Pass** | Method matches business maturity | **Fail** | Gordon on high-growth tech without bounds | **Difficulty** | L2 | **Reward** | L2 |

Pass criteria: Method matches business maturity | **Fail** | Gordon on high-growth tech without bounds | **Difficulty** | L2 | **Reward** | L2 |

Fail conditions: Gordon on high-growth tech without bounds | **Difficulty** | L2 | **Reward** | L2 |

Difficulty: L2 | **Reward** | L2 |

DCF-007: Bound Terminal Growth Rate

Description: Terminal $g \leq$ long-term GDP/industry growth with justification. | **Dependencies** | DCF-006, IND-002 | **Pass** | g bounded and cited | **Fail** | $g >$ GDP without moat case (**mandate fail**) | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: DCF-006, IND-002 | **Pass** | g bounded and cited | **Fail** | $g >$ GDP without moat case (**mandate fail**) | **Difficulty** | L3 |

Pass criteria: g bounded and cited | **Fail** | $g >$ GDP without moat case (**mandate fail**) | **Difficulty** | L3 |

Fail conditions: $g >$ GDP without moat case (**mandate fail**) | **Difficulty** | L3 |

Difficulty: L3 |

DCF-008: Compute Gordon Terminal Value

Description: $TV = UFCF(n+1) / (WACC - g)$. | **Dependencies** | DCF-005, DCF-007 | **Pass** | TV calculation verified in Python | **Fail** | $g \geq WACC$ | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: DCF-005, DCF-007 | **Pass** | TV calculation verified in Python | **Fail** | $g \geq WACC$ | **Difficulty** |

L3 |

Pass criteria: TV calculation verified in Python | **Fail** | $g \geq WACC$ | **Difficulty** | L3 |

Fail conditions: $g \geq WACC$ | **Difficulty** | L3 |

Difficulty: L3 |

DCF-009: Compute Exit Multiple Terminal Value

Description: $TV = \text{Exit EBITDA} \times \text{peer median multiple}$. | **Dependencies** | COMP-006, M3-004 | **Pass** | Multiple from comp set | **Fail** | Arbitrary multiple | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: COMP-006, M3-004 | **Pass** | Multiple from comp set | **Fail** | Arbitrary multiple | **Difficulty** | L3 |

Pass criteria: Multiple from comp set | **Fail** | Arbitrary multiple | **Difficulty** | L3 |

Fail conditions: Arbitrary multiple | **Difficulty** | L3 |

Difficulty: L3 |

DCF-010: Discount UFCF and TV to Present Value

Description: $PV \text{ of explicit period} + PV \text{ of TV}$. | **Dependencies** | DCF-005, DCF-008 or DCF-009 | **Pass** | Mid-year convention documented | **Fail** | $TV > 80\%$ without sensitivity flag | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: DCF-005, DCF-008 or DCF-009 | **Pass** | Mid-year convention documented | **Fail** | $TV > 80\%$ without sensitivity flag | **Difficulty** | L3 |

Pass criteria: Mid-year convention documented | **Fail** | $TV > 80\%$ without sensitivity flag | **Difficulty** | L3 |

Fail conditions: $TV > 80\%$ without sensitivity flag | **Difficulty** | L3 |

Difficulty: L3 |

DCF-011: Bridge EV to Equity Value

Description: $\text{Equity} = EV - \text{net debt} + \text{non-operating assets}$. | **Dependencies** | DCF-010, D-019 | **Pass** | Net debt matches D-019 | **Fail** | Omits minority interest | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-010, D-019 | **Pass** | Net debt matches D-019 | **Fail** | Omits minority interest | **Difficulty** | L2 |

Pass criteria: Net debt matches D-019 | **Fail** | Omits minority interest | **Difficulty** | L2 |

Fail conditions: Omits minority interest | **Difficulty** | L2 |

Difficulty: L2 |

DCF-012: Compute Implied Share Price

Description: $\text{Equity value} \div \text{diluted shares}$. | **Dependencies** | DCF-011, D-018 | **Pass** | Uses diluted shares | **Fail** | Basic shares for dilutive co | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-011, D-018 | **Pass** | Uses diluted shares | **Fail** | Basic shares for dilutive co | **Difficulty** | L2 |

Pass criteria: Uses diluted shares | **Fail** | Basic shares for dilutive co | **Difficulty** | L2 |

Fail conditions: Basic shares for dilutive co | **Difficulty** | L2 |

Difficulty: L2 |

DCF-013: WACC x Terminal g Sensitivity Table

Description: 5x5 grid of implied prices. | **Dependencies** | DCF-004, DCF-007, DCF-012 | **Pass** | Grid mathematically consistent | **Fail** | Single point estimate only | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-004, DCF-007, DCF-012 | **Pass** | Grid mathematically consistent | **Fail** | Single point estimate only | **Difficulty** | L2 |

Pass criteria: Grid mathematically consistent | **Fail** | Single point estimate only | **Difficulty** | L2 |

Fail conditions: Single point estimate only | **Difficulty** | L2 |

Difficulty: L2 |

DCF-014: Exit Multiple Sensitivity

Description: Sensitivity of price to exit EV/EBITDA +/-2 turns. | **Dependencies** | DCF-009 | **Pass** | Range presented | **Fail** | No sensitivity | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-009 | **Pass** | Range presented | **Fail** | No sensitivity | **Difficulty** | L2 |

Pass criteria: Range presented | **Fail** | No sensitivity | **Difficulty** | L2 |

Fail conditions: No sensitivity | **Difficulty** | L2 |

Difficulty: L2 |

DCF-015: Compare DCF to Current Market Price

Description: Compute upside/downside %. | **Dependencies** | DCF-012, market price | **Pass** | % calc correct | **Fail** | Stale price without date | **Difficulty** | L1 |

Inputs: -

Outputs: -

Dependencies: DCF-012, market price | **Pass** | % calc correct | **Fail** | Stale price without date | **Difficulty** | L1 |

Pass criteria: % calc correct | **Fail** | Stale price without date | **Difficulty** | L1 |

Fail conditions: Stale price without date | **Difficulty** | L1 |

Difficulty: L1 |

DCF-016: Document All DCF Assumptions

Description: Audit trail for WACC, g, margins, CapEx. | **Dependencies** | DCF-001-015 | **Pass** | All assumptions cited | **Fail** | Missing source | **Difficulty** | L2 | **Reward** | L3 |

Inputs: -

Outputs: -

Dependencies: DCF-001-015 | **Pass** | All assumptions cited | **Fail** | Missing source | **Difficulty** | L2 | **Reward** | L3 |

Pass criteria: All assumptions cited | **Fail** | Missing source | **Difficulty** | L2 | **Reward** | L3 |

Fail conditions: Missing source | **Difficulty** | L2 | **Reward** | L3 |

Difficulty: L2 | **Reward** | L3 |

DCF-017: Flag Unverified DCF Inputs

Description: Mark missing data; do not interpolate. | **Dependencies** | DCF-016 | **Pass** | Gaps flagged as unverified | **Fail** | Hallucinated input (**critical L3**) | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: DCF-016 | **Pass** | Gaps flagged as unverified | **Fail** | Hallucinated input (**critical L3**) | **Difficulty** | L2 |

Pass criteria: Gaps flagged as unverified | **Fail** | Hallucinated input (**critical L3**) | **Difficulty** | L2 |

Fail conditions: Hallucinated input (**critical L3**) | **Difficulty** | L2 |

Difficulty: L2 |

Layer 2C - Trading Comps

COMP-001: Select 5-8 Peers with Rationale

Description: Select 5-8 Peers with Rationale

Inputs: -

Outputs: -

Dependencies: D-015 | **Pass** | Sector-appropriate list | **Fail** | Absurd peer (e.g., NVDA for KO) | **Difficulty** | L2 |

Pass criteria: Sector-appropriate list | **Fail** | Absurd peer (e.g., NVDA for KO) | **Difficulty** | L2 |

Fail conditions: Absurd peer (e.g., NVDA for KO) | **Difficulty** | L2 |

Difficulty: L2 |

COMP-002: Pull LTM Revenue, EBITDA, EPS for Peers

Description: Pull LTM Revenue, EBITDA, EPS for Peers

Inputs: -

Outputs: -

Dependencies: D-016, D-006 | **Pass** | Same LTM period all peers | **Fail** | Mixed fiscal periods | **Difficulty** | L2 |

Pass criteria: Same LTM period all peers | ****Fail**** | Mixed fiscal periods | ****Difficulty**** | L2 |

Fail conditions: Mixed fiscal periods | ****Difficulty**** | L2 |

Difficulty: L2 |

COMP-003: Normalize Peer EBITDA

Description: Normalize Peer EBITDA

Inputs: -

Outputs: -

Dependencies: COMP-002, FS-003 | ****Pass**** | One-times added back with citation | ****Fail**** | Raw GAAP only | ****Difficulty**** | L3 |

Pass criteria: One-times added back with citation | ****Fail**** | Raw GAAP only | ****Difficulty**** | L3 |

Fail conditions: Raw GAAP only | ****Difficulty**** | L3 |

Difficulty: L3 |

COMP-004: Calculate Peer Enterprise Values

Description: Calculate Peer Enterprise Values

Inputs: -

Outputs: -

Dependencies: COMP-002, market data, D-019 | ****Pass**** | $EV = MC + \text{net debt}$ | ****Fail**** | Equity value used as EV | ****Difficulty**** | L2 |

Pass criteria: $EV = MC + \text{net debt}$ | ****Fail**** | Equity value used as EV | ****Difficulty**** | L2 |

Fail conditions: Equity value used as EV | ****Difficulty**** | L2 |

Difficulty: L2 |

COMP-005: Compute Peer Multiples

Description: EV/EBITDA, EV/Revenue, P/E, P/FCF. | ****Dependencies**** | COMP-003, COMP-004 | ****Pass**** | Median and mean calculated | ****Fail**** | Division by zero unhandled | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: COMP-003, COMP-004 | ****Pass**** | Median and mean calculated | ****Fail**** | Division by zero unhandled | ****Difficulty**** | L2 |

Pass criteria: Median and mean calculated | ****Fail**** | Division by zero unhandled | ****Difficulty**** | L2 |

Fail conditions: Division by zero unhandled | ****Difficulty**** | L2 |

Difficulty: L2 |

COMP-006: Summarize Multiple Statistics

Description: Mean, median, 25th/75th percentile. | ****Dependencies**** | COMP-005 | ****Pass**** | Stats correct | ****Fail**** | Outlier not flagged | ****Difficulty**** | L2 |

Inputs: -

Outputs: -

Dependencies: COMP-005 | ****Pass**** | Stats correct | ****Fail**** | Outlier not flagged | ****Difficulty**** | L2 |

Pass criteria: Stats correct | ****Fail**** | Outlier not flagged | ****Difficulty**** | L2 |

Fail conditions: Outlier not flagged | ****Difficulty**** | L2 |

Difficulty: L2 |

COMP-007: Apply Median EV/EBITDA to Target NTM EBITDA

Description: Apply Median EV/EBITDA to Target NTM EBITDA

Inputs: -

Outputs: -

Dependencies: COMP-006, M3-004 | ****Pass**** | Implied EV reasonable | ****Fail**** | NTM EBITDA not from model | ****Difficulty**** | L3 |

Pass criteria: Implied EV reasonable | ****Fail**** | NTM EBITDA not from model | ****Difficulty**** | L3 |

Fail conditions: NTM EBITDA not from model | ****Difficulty**** | L3 |

Difficulty: L3 |

COMP-008: Apply Median P/E to Target NTM EPS

Description: Apply Median P/E to Target NTM EPS

Inputs: -

Outputs: -

Dependencies: COMP-006, M3-001 | ****Pass**** | Implied price calculated | ****Fail**** | Uses GAAP EPS with one-time | ****Difficulty**** | L3 |

Pass criteria: Implied price calculated | ****Fail**** | Uses GAAP EPS with one-time | ****Difficulty**** | L3 |

Fail conditions: Uses GAAP EPS with one-time | ****Difficulty**** | L3 |

Difficulty: L3 |

COMP-009: Explain Premium/Discount vs Peers

Description: Explain Premium/Discount vs Peers

Inputs: -

Outputs: -

Dependencies: COMP-007, COMP-008 | ****Pass**** | 3+ factor rationale | ****Fail**** | "Deserves premium" without evidence | ****Difficulty**** | L2 | ****Reward**** | L2 |

Pass criteria: 3+ factor rationale | ****Fail**** | "Deserves premium" without evidence | ****Difficulty**** | L2 | ****Reward**** | L2 |

Fail conditions: "Deserves premium" without evidence | ****Difficulty**** | L2 | ****Reward**** | L2 |

Difficulty: L2 | ****Reward**** | L2 |

COMP-010: Sector-Specific Multiple (EV/Sub, EV/Sub)

Description: Sector-Specific Multiple (EV/Sub, EV/Sub)

Inputs: -

Outputs: -

Dependencies: COMP-005, D-014 | ****Pass**** | Sector metric correctly computed | ****Fail**** | Wrong denominator | ****Difficulty**** | L3 | ****Sectors**** | NFLX, SPOT, DIS |

Pass criteria: Sector metric correctly computed | ****Fail**** | Wrong denominator | ****Difficulty**** | L3 | ****Sectors**** | NFLX, SPOT, DIS |

Fail conditions: Wrong denominator | ****Difficulty**** | L3 | ****Sectors**** | NFLX, SPOT, DIS |

Difficulty: L3 | ****Sectors**** | NFLX, SPOT, DIS |

COMP-011: Flag Inappropriate Comp

Description: Identify and reject wrong-sector peer inclusion. | ****Dependencies**** | COMP-001 | ****Pass**** | Flags bad comp | ****Fail**** | Accepts all peers uncritically | ****Difficulty**** | L2 | ****Reward**** | L2 |

Inputs: -

Outputs: -

Dependencies: COMP-001 | ****Pass**** | Flags bad comp | ****Fail**** | Accepts all peers uncritically | ****Difficulty**** | L2 | ****Reward**** | L2 |

Pass criteria: Flags bad comp | ****Fail**** | Accepts all peers uncritically | ****Difficulty**** | L2 | ****Reward**** | L2 |

Fail conditions: Accepts all peers uncritically | ****Difficulty**** | L2 | ****Reward**** | L2 |

Difficulty: L2 | ****Reward**** | L2 |

COMP-012: Refresh Comps Post-Earnings

Description: Refresh Comps Post-Earnings

Inputs: -

Outputs: -

Dependencies: COMP-001-011, EARN-008 | ****Pass**** | Updated LTM/NTM | ****Fail**** | Stale peer data | ****Difficulty**** | L2 |

Pass criteria: Updated LTM/NTM | ****Fail**** | Stale peer data | ****Difficulty**** | L2 |

Fail conditions: Stale peer data | ****Difficulty**** | L2 |

Difficulty: L2 |

Layer 2D - LBO Model

LBO-001: Set Entry EV/Multiple

Description: Set Entry EV/Multiple

Inputs: -

Outputs: -

Dependencies: COMP-006 or market | ****Pass**** | Entry rationale documented | ****Fail**** | Arbitrary entry | ****Difficulty**** | L2 |

Pass criteria: Entry rationale documented | ****Fail**** | Arbitrary entry | ****Difficulty**** | L2 |

Fail conditions: Arbitrary entry | ****Difficulty**** | L2 |

Difficulty: L2 |

LBO-002: Build Sources & Uses

Description: Build Sources & Uses

Inputs: -

Outputs: -

Dependencies: LBO-001 | **Pass** | S&U balance | **Fail** | S?U | **Difficulty** | L2 |

Pass criteria: S&U balance | **Fail** | S?U | **Difficulty** | L2 |

Fail conditions: S?U | **Difficulty** | L2 |

Difficulty: L2 |

LBO-003: Layer Debt Tranches

Description: Senior + sub at market rates; 60-70% leverage. | **Dependencies** | LBO-002 | **Pass** | Leverage stated | **Fail** | 90% leverage on cyclical without flag | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: LBO-002 | **Pass** | Leverage stated | **Fail** | 90% leverage on cyclical without flag | **Difficulty** | L3 |

Pass criteria: Leverage stated | **Fail** | 90% leverage on cyclical without flag | **Difficulty** | L3 |

Fail conditions: 90% leverage on cyclical without flag | **Difficulty** | L3 |

Difficulty: L3 |

LBO-004: Project 5-Year FCF Under PE Ownership

Description: Project 5-Year FCF Under PE Ownership

Inputs: -

Outputs: -

Dependencies: M3-017, LBO-003 | **Pass** | FCF from model not reinvented | **Fail** | Ignores capex maintenance | **Difficulty** | L3 |

Pass criteria: FCF from model not reinvented | **Fail** | Ignores capex maintenance | **Difficulty** | L3 |

Fail conditions: Ignores capex maintenance | **Difficulty** | L3 |

Difficulty: L3 |

LBO-005: Model Debt Paydown from FCF

Description: Model Debt Paydown from FCF

Inputs: -

Outputs: -

Dependencies: LBO-004 | **Pass** | Debt schedule ties | **Fail** | Interest not cash swept | **Difficulty** | L3 |

Pass criteria: Debt schedule ties | **Fail** | Interest not cash swept | **Difficulty** | L3 |

Fail conditions: Interest not cash swept | **Difficulty** | L3 |

Difficulty: L3 |

LBO-006: Set Exit EV/EBITDA Year 5

Description: Set Exit EV/EBITDA Year 5

Inputs: -

Outputs: -

Dependencies: COMP-006 | **Pass** | Exit multiple bounded | **Fail** | Exit > entry without growth case | **Difficulty** | L2 |

Pass criteria: Exit multiple bounded | **Fail** | Exit > entry without growth case | **Difficulty** | L2 |

Fail conditions: Exit > entry without growth case | **Difficulty** | L2 |

Difficulty: L2 |

LBO-007: Calculate IRR and MOIC

Description: Calculate IRR and MOIC

Inputs: -

Outputs: -

Dependencies: LBO-005, LBO-006 | **Pass** | IRR verified in Python | **Fail** | Manual IRR error | **Difficulty** | L3 |

Pass criteria: IRR verified in Python | **Fail** | Manual IRR error | **Difficulty** | L3 |

Fail conditions: Manual IRR error | **Difficulty** | L3 |

Difficulty: L3 |

LBO-008: Entry vs Exit Multiple Sensitivity

Description: Entry vs Exit Multiple Sensitivity

Inputs: -

Outputs: -

Dependencies: LBO-007 | **Pass** | IRR grid produced | **Fail** | Single scenario | **Difficulty** | L2 |

Pass criteria: IRR grid produced | **Fail** | Single scenario | **Difficulty** | L2 |

Fail conditions: Single scenario | **Difficulty** | L2 |

Difficulty: L2 |

LBO-009: Assess LBO Feasibility

Description: FCF stability, capex intensity, cyclical assessment. | **Dependencies** | LBO-007, M3-017 | **Pass** | Feasibility memo with constraints | **Fail** | Declares LBO on negative FCF name | **Difficulty** | L2 | **Sectors** | PEP, MCD, CMCSA |

Inputs: -

Outputs: -

Dependencies: LBO-007, M3-017 | **Pass** | Feasibility memo with constraints | **Fail** | Declares LBO on negative FCF name | **Difficulty** | L2 | **Sectors** | PEP, MCD, CMCSA |

Pass criteria: Feasibility memo with constraints | **Fail** | Declares LBO on negative FCF name | **Difficulty** | L2 | **Sectors** | PEP, MCD, CMCSA |

Fail conditions: Declares LBO on negative FCF name | **Difficulty** | L2 | **Sectors** | PEP, MCD, CMCSA |

Difficulty: L2 | **Sectors** | PEP, MCD, CMCSA |

LBO-010: Compare LBO Implied Price to DCF

Description: Compare LBO Implied Price to DCF

Inputs: -

Outputs: -

Dependencies: LBO-007, DCF-012 | **Pass** | Triangulation comment | **Fail** | Ignores 50% dispersion | **Difficulty** | L2 |

Pass criteria: Triangulation comment | **Fail** | Ignores 50% dispersion | **Difficulty** | L2 |

Fail conditions: Ignores 50% dispersion | **Difficulty** | L2 |

Difficulty: L2 |

Layer 2E - Sum-of-the-Parts

SOTP-001: Map Segments to Valuation Buckets

Description: Map Segments to Valuation Buckets

Inputs: -

Outputs: -

Dependencies: segment note | **Pass** | All reportable segments mapped | **Fail** | Missing segment | **Difficulty** | L2 | **Sectors** | GOOGL, AMZN, DIS, CMCSA |

Pass criteria: All reportable segments mapped | **Fail** | Missing segment | **Difficulty** | L2 | **Sectors** | GOOGL, AMZN, DIS, CMCSA |

Fail conditions: Missing segment | **Difficulty** | L2 | **Sectors** | GOOGL, AMZN, DIS, CMCSA |

Difficulty: L2 | **Sectors** | GOOGL, AMZN, DIS, CMCSA |

SOTP-002: Assign Peer Set per Segment

Description: Assign Peer Set per Segment

Inputs: -

Outputs: -

Dependencies: SOTP-001, D-015 | **Pass** | Distinct peers per segment | **Fail** | Same comp set for cloud and retail | **Difficulty** | L3 |

Pass criteria: Distinct peers per segment | **Fail** | Same comp set for cloud and retail | **Difficulty** | L3 |

Fail conditions: Same comp set for cloud and retail | **Difficulty** | L3 |

Difficulty: L3 |

SOTP-003: Value Segment A (EV/EBITDA)

Description: Value Segment A (EV/EBITDA)

Inputs: -

Outputs: -

Dependencies: SOTP-002, segment financials | ****Pass**** | Segment EBITDA normalized | ****Fail**** | Unallocated costs ignored | ****Difficulty**** | L3 |

Pass criteria: Segment EBITDA normalized | ****Fail**** | Unallocated costs ignored | ****Difficulty**** | L3 |

Fail conditions: Unallocated costs ignored | ****Difficulty**** | L3 |

Difficulty: L3 |

SOTP-004: Value Segment B (Different Multiple)

Description: Value Segment B (Different Multiple)

Inputs: -

Outputs: -

Dependencies: SOTP-002 | ****Pass**** | Multiple appropriate to segment growth | ****Fail**** | Single multiple all segments | ****Difficulty**** | L3 |

Pass criteria: Multiple appropriate to segment growth | ****Fail**** | Single multiple all segments | ****Difficulty**** | L3 |

Fail conditions: Single multiple all segments | ****Difficulty**** | L3 |

Difficulty: L3 |

SOTP-005: Apply Holdco Discount

Description: Apply Holdco Discount

Inputs: -

Outputs: -

Dependencies: SOTP-003, SOTP-004 | ****Pass**** | Discount 0-25% with rationale | ****Fail**** | No discount on unrelated segments | ****Difficulty**** | L2 | ****Reward**** | L2 |

Pass criteria: Discount 0-25% with rationale | ****Fail**** | No discount on unrelated segments | ****Difficulty**** | L2 | ****Reward**** | L2 |

Fail conditions: No discount on unrelated segments | ****Difficulty**** | L2 | ****Reward**** | L2 |

Difficulty: L2 | ****Reward**** | L2 |

SOTP-006: Allocate Corporate Overhead

Description: Allocate Corporate Overhead

Inputs: -

Outputs: -

Dependencies: FS-023 | ****Pass**** | Overhead removed from segments | ****Fail**** | Double-counted | ****Difficulty**** | L3 |

Pass criteria: Overhead removed from segments | ****Fail**** | Double-counted | ****Difficulty**** | L3 |

Fail conditions: Double-counted | ****Difficulty**** | L3 |

Difficulty: L3 |

SOTP-007: Sum to EV -> Equity -> Per Share

Description: Sum to EV -> Equity -> Per Share

Inputs: -

Outputs: -

Dependencies: SOTP-003-006, D-019 | ****Pass**** | Arithmetic correct | ****Fail**** | Net debt omitted | ****Difficulty**** | L2 |

Pass criteria: Arithmetic correct | ****Fail**** | Net debt omitted | ****Difficulty**** | L2 |

Fail conditions: Net debt omitted | ****Difficulty**** | L2 |

Difficulty: L2 |

SOTP-008: Compare SOTP to Consolidated DCF

Description: Compare SOTP to Consolidated DCF

Inputs: -

Outputs: -

Dependencies: SOTP-007, DCF-012 | ****Pass**** | Explains premium/discount | ****Fail**** | Ignores divergence | ****Difficulty**** | L2 |

Pass criteria: Explains premium/discount | ****Fail**** | Ignores divergence | ****Difficulty**** | L2 |

Fail conditions: Ignores divergence | ****Difficulty**** | L2 |

Difficulty: L2 |

SOTP-009: Flag Hidden / Unlisted Value

Description: Identify undisclosed subsidiaries (Waymo, etc.). | ****Dependencies**** | SOTP-001 | ****Pass**** | Flags unmodeled value | ****Fail**** | Assigns value without disclosure | ****Difficulty**** | L2 | ****Reward**** | L3 |

Inputs: -

Outputs: -

Dependencies: SOTP-001 | **Pass** | Flags unmodeled value | **Fail** | Assigns value without disclosure | **Difficulty** | L2 | **Reward** | L3 |

Pass criteria: Flags unmodeled value | **Fail** | Assigns value without disclosure | **Difficulty** | L2 | **Reward** | L3 |

Fail conditions: Assigns value without disclosure | **Difficulty** | L2 | **Reward** | L3 |

Difficulty: L2 | **Reward** | L3 |

Layer 2F - Dividend Discount Model

DDM-001: Extract 10-Year Dividend History

Description: Extract 10-Year Dividend History

Inputs: -

Outputs: -

Dependencies: CF, dividend footnote | **Pass** | DPS series correct | **Fail** | Special dividend not flagged | **Difficulty** | L1 | **Sectors** | PEP, KO, MCD |

Pass criteria: DPS series correct | **Fail** | Special dividend not flagged | **Difficulty** | L1 | **Sectors** | PEP, KO, MCD |

Fail conditions: Special dividend not flagged | **Difficulty** | L1 | **Sectors** | PEP, KO, MCD |

Difficulty: L1 | **Sectors** | PEP, KO, MCD |

DDM-002: Compute Payout vs FCF and EPS Coverage

Description: Compute Payout vs FCF and EPS Coverage

Inputs: -

Outputs: -

Dependencies: DDM-001, M3-014, M3-017 | **Pass** | Coverage ratios calculated | **Fail** | Uses EPS only ignoring FCF | **Difficulty** | L2 |

Pass criteria: Coverage ratios calculated | **Fail** | Uses EPS only ignoring FCF | **Difficulty** | L2 |

Fail conditions: Uses EPS only ignoring FCF | **Difficulty** | L2 |

Difficulty: L2 |

DDM-003: Project Dividend Growth 5 Years

Description: Project Dividend Growth 5 Years

Inputs: -

Outputs: -

Dependencies: DDM-001, M3-014 | **Pass** | Growth bounded vs history | **Fail** | 15% DPS growth forever | **Difficulty** | L2 |

Pass criteria: Growth bounded vs history | **Fail** | 15% DPS growth forever | **Difficulty** | L2 |

Fail conditions: 15% DPS growth forever | **Difficulty** | L2 |

Difficulty: L2 |

DDM-004: Calculate Cost of Equity (CAPM)

Description: Calculate Cost of Equity (CAPM)

Inputs: -

Outputs: -

Dependencies: DCF-001 | **Pass** | Same Ke methodology as DCF | **Fail** | Different Ke unexplained | **Difficulty** | L2 |

Pass criteria: Same Ke methodology as DCF | **Fail** | Different Ke unexplained | **Difficulty** | L2 |

Fail conditions: Different Ke unexplained | **Difficulty** | L2 |

Difficulty: L2 |

DDM-005: Run Gordon or Multi-Stage DDM

Description: Run Gordon or Multi-Stage DDM

Inputs: -

Outputs: -

Dependencies: DDM-003, DDM-004 | **Pass** | Intrinsic value calculated | **Fail** | $g \geq Ke$ | **Difficulty** | L3 |

Pass criteria: Intrinsic value calculated | **Fail** | $g \geq Ke$ | **Difficulty** | L3 |

Fail conditions: $g \geq Ke$ | **Difficulty** | L3 |

Difficulty: L3 |

DDM-006: Dividend Sustainability Stress Test

Description: Model -20% FCF scenario; can dividend be maintained? | **Dependencies** | DDM-002, M3-018 | **Pass** | Stress outcome stated | **Fail** | Ignores payout risk (**mandate**) | **Difficulty** | L3 |

Inputs: -

Outputs: -

Dependencies: DDM-002, M3-018 | **Pass** | Stress outcome stated | **Fail** | Ignores payout risk (**mandate**) | **Difficulty** | L3 |

Pass criteria: Stress outcome stated | **Fail** | Ignores payout risk (**mandate**) | **Difficulty** | L3 |

Fail conditions: Ignores payout risk (**mandate**) | **Difficulty** | L3 |

Difficulty: L3 |

DDM-007: Compare Yield to Peer Income Names

Description: Compare Yield to Peer Income Names

Inputs: -

Outputs: -

Dependencies: DDM-001, COMP-001 | **Pass** | Relative yield table | **Fail** | Wrong peer set | **Difficulty** | L2 |

Pass criteria: Relative yield table | **Fail** | Wrong peer set | **Difficulty** | L2 |

Fail conditions: Wrong peer set | **Difficulty** | L2 |

Difficulty: L2 |

Layer 2G - M&A Accretion/Dilution

MA-001: Model Pro Forma Combined IS

Description: Model Pro Forma Combined IS

Inputs: -

Outputs: -

Dependencies: M3-001, target IS | **Pass** | Combination mechanics correct | **Fail** | Double-counts revenue | **Difficulty** | L3 | **Sectors** | Media M&A |

Pass criteria: Combination mechanics correct | **Fail** | Double-counts revenue | **Difficulty** | L3 | **Sectors** | Media M&A |

Fail conditions: Double-counts revenue | **Difficulty** | L3 | **Sectors** | Media M&A |

Difficulty: L3 | **Sectors** | Media M&A |

MA-002: Synergy Assumptions with Timing

Description: Synergy Assumptions with Timing

Inputs: -

Outputs: -

Dependencies: MA-001 | **Pass** | Synergies phased | **Fail** | Day-1 full synergies | **Difficulty** | L3 |

Pass criteria: Synergies phased | **Fail** | Day-1 full synergies | **Difficulty** | L3 |

Fail conditions: Day-1 full synergies | **Difficulty** | L3 |

Difficulty: L3 |

MA-003: Compute EPS Accretion/Dilution

Description: Compute EPS Accretion/Dilution

Inputs: -

Outputs: -

Dependencies: MA-001, MA-002 | **Pass** | Accretion % in Y1 and Y2 | **Fail** | Wrong share count | **Difficulty** | L3 |

Pass criteria: Accretion % in Y1 and Y2 | **Fail** | Wrong share count | **Difficulty** | L3 |

Fail conditions: Wrong share count | **Difficulty** | L3 |

Difficulty: L3 |

MA-004: Deal Sources & Uses / Financing

Description: Deal Sources & Uses / Financing

Inputs: -

Outputs: -

Dependencies: MA-003 | **Pass** | S&U balances | **Fail** | Unfunded purchase price | **Difficulty** | L3 |

Pass criteria: S&U balances | **Fail** | Unfunded purchase price | **Difficulty** | L3 |

Fail conditions: Unfunded purchase price | **Difficulty** | L3 |

Difficulty: L3 |

Layer 3 - Valuation Synthesis

VAL-001: Build Football Field Chart

Description: Combine DCF, comps, SOTP, LBO ranges on one chart. | **Dependencies** | ≥ 2 of DCF-012, COMP-007, SOTP-007, LBO-007 | **Pass** | All methods shown with ranges | **Fail** | Single method only | **Difficulty** | L2 |

Inputs: -

Outputs: -

Dependencies: ≥ 2 of DCF-012, COMP-007, SOTP-007, LBO-007 | **Pass** | All methods shown with ranges | **Fail** | Single method only | **Difficulty** | L2 |

Pass criteria: All methods shown with ranges | **Fail** | Single method only | **Difficulty** | L2 |

Fail conditions: Single method only | **Difficulty** | L2 |

Difficulty: L2 |

VAL-002: Assign Methodology Weights

Description: Weight each method for blended target. | **Dependencies** | VAL-001 | **Pass** | Weights sum to 100%; rationale | **Fail** | Equal weights without thought | **Difficulty** | L2 | **Reward** | L2 |

Inputs: -

Outputs: -

Dependencies: VAL-001 | **Pass** | Weights sum to 100%; rationale | **Fail** | Equal weights without thought | **Difficulty** | L2 | **Reward** | L2 |

Pass criteria: Weights sum to 100%; rationale | **Fail** | Equal weights without thought | **Difficulty** | L2 | **Reward** | L2 |

Fail conditions: Equal weights without thought | **Difficulty** | L2 | **Reward** | L2 |

Difficulty: L2 | **Reward** | L2 |

VAL-003: Set 12-Month Price Target with Range

Description: Set 12-Month Price Target with Range

Inputs: -

Outputs: -

Dependencies: VAL-002 | **Pass** | Bull/base/bear stated | **Fail** | Point target only | **Difficulty** | L2 |

Pass criteria: Bull/base/bear stated | **Fail** | Point target only | **Difficulty** | L2 |

Fail conditions: Point target only | **Difficulty** | L2 |

Difficulty: L2 |

VAL-004: Compute Upside/Downside to Current

Description: Compute Upside/Downside to Current

Inputs: -

Outputs: -

Dependencies: VAL-003, market price | **Pass** | Return calc with date | **Fail** | Wrong denominator | **Difficulty** | L1 |

Pass criteria: Return calc with date | **Fail** | Wrong denominator | **Difficulty** | L1 |

Fail conditions: Wrong denominator | **Difficulty** | L1 |

Difficulty: L1 |

VAL-005: Reconcile Wide Valuation Dispersion

Description: Explain if methods diverge $>30\%$. | **Dependencies** | VAL-001 | **Pass** | Identifies driver of spread | **Fail** | Ignores dispersion | **Difficulty** | L2 | **Reward** | L2 |

Inputs: -

Outputs: -

Dependencies: VAL-001 | **Pass** | Identifies driver of spread | **Fail** | Ignores dispersion | **Difficulty** | L2 | **Reward** | L2 |

Pass criteria: Identifies driver of spread | **Fail** | Ignores dispersion | **Difficulty** | L2 | **Reward** | L2 |

Fail conditions: Ignores dispersion | **Difficulty** | L2 | **Reward** | L2 |

Difficulty: L2 | **Reward** | L2 |

VAL-006: Update Target Post-Model Refresh

Description: Update Target Post-Model Refresh

Inputs: -

Outputs: -

Dependencies: VAL-003, EARN-008 | **Pass** | Revision tied to model change | **Fail** | PT unchanged despite 10% EPS change | **Difficulty** | L2 |

Pass criteria: Revision tied to model change | **Fail** | PT unchanged despite 10% EPS change | **Difficulty** | L2 |

Fail conditions: PT unchanged despite 10% EPS change | **Difficulty** | L2 |

Difficulty: L2 |

Layer 4 - Industry Analysis

IND-001: Porter's Five Forces Analysis

Description: Porter's Five Forces Analysis

Inputs: -

Outputs: Forces memo with evidence | **Difficulty** | L2 | **Reward** | L2 |

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 | **Reward** | L2 |

IND-002: TAM/SAM/SOM Sizing

Description: TAM/SAM/SOM Sizing

Inputs: IR decks, industry reports (bounded) | **Pass** | Ranges not point estimates | **Difficulty** | L2 |

Outputs: -

Dependencies: -

Pass criteria: Ranges not point estimates | **Difficulty** | L2 |

Fail conditions: -

Difficulty: L2 |

IND-003: Market Share Trends

Description: Market Share Trends

Inputs: -

Outputs: -

Dependencies: D-016, D-006 | **Pass** | Share calc methodology stated | **Difficulty** | L2 |

Pass criteria: Share calc methodology stated | **Difficulty** | L2 |

Fail conditions: -

Difficulty: L2 |

IND-004: Pricing Power Analysis

Description: Pricing Power Analysis

Inputs: -

Outputs: -

Dependencies: FS-013, margin trends | **Pass** | Links margins to pricing | **Difficulty** | L2 | **Sectors** | Consumer |

Pass criteria: Links margins to pricing | **Difficulty** | L2 | **Sectors** | Consumer |

Fail conditions: -

Difficulty: L2 | **Sectors** | Consumer |

IND-005: Regulatory Overlay

Description: Regulatory Overlay

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Identifies sector-specific regulation | **Difficulty** | L2 | **Sectors** | Tech antitrust, media content regs |

Fail conditions: -

Difficulty: L2 | **Sectors** | Tech antitrust, media content regs |

IND-006: Disruption Risk Matrix

Description: Disruption Risk Matrix

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: AI/streaming/DTC threats assessed | ****Difficulty**** | L2 | ****Sectors**** | Tech, Media |

Fail conditions: -

Difficulty: L2 | ****Sectors**** | Tech, Media |

IND-007: Commodity / Input Cost Sensitivity

Description: Commodity / Input Cost Sensitivity

Inputs: -

Outputs: -

Dependencies: FRED commodities, M3-005 | ****Pass**** | Quantified sensitivity | ****Difficulty**** | L3 | ****Sectors**** | PEP, KO, MDLZ |

Pass criteria: Quantified sensitivity | ****Difficulty**** | L3 | ****Sectors**** | PEP, KO, MDLZ |

Fail conditions: -

Difficulty: L3 | ****Sectors**** | PEP, KO, MDLZ |

IND-008: Capital Intensity / ROIC vs Peers

Description: Capital Intensity / ROIC vs Peers

Inputs: -

Outputs: -

Dependencies: FS-024, COMP-001 | ****Pass**** | ROIC comparison table | ****Difficulty**** | L3 |

Pass criteria: ROIC comparison table | ****Difficulty**** | L3 |

Fail conditions: -

Difficulty: L3 |

IND-009: Industry Cycle Positioning

Description: Industry Cycle Positioning

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Early/mid/late cycle call with evidence | ****Difficulty**** | L2 | ****Reward**** | L2 |

Fail conditions: -

Difficulty: L2 | ****Reward**** | L2 |

Layer 5 - Earnings Workflow

EARN-001: Pre-Earnings Preview Note

Description: Pre-Earnings Preview Note

Inputs: -

Outputs: Expected KPIs, key debates | ****Difficulty**** | L2 |

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

EARN-002: Extract Consensus Expectations

Description: Extract Consensus Expectations

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Source cited or flagged unavailable | ****Fail**** | Fabricated consensus | ****Difficulty**** | L2 | ****Reward**** | L3 |

Fail conditions: Fabricated consensus | ****Difficulty**** | L2 | ****Reward**** | L3 |

Difficulty: L2 | ****Reward**** | L3 |

EARN-003: Identify Key KPIs to Watch

Description: Identify Key KPIs to Watch

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

EARN-004: Real-Time Earnings Recap

Description: Real-Time Earnings Recap

Inputs: -

Outputs: Headline beats/misses | ****Difficulty**** | L2 |

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

EARN-005: Compare Actual vs Preview

Description: Compare Actual vs Preview

Inputs: -

Outputs: -

Dependencies: EARN-001, EARN-004 | ****Pass**** | Variance explained | ****Difficulty**** | L2 |

Pass criteria: Variance explained | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

EARN-006: Extract Guidance Raise/Lower/Maintain

Description: Extract Guidance Raise/Lower/Maintain

Inputs: -

Outputs: -

Dependencies: D-011 | ****Pass**** | Direction correct with quote | ****Difficulty**** | L2 |

Pass criteria: Direction correct with quote | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

EARN-007: Guidance Drift - Call vs Subsequent 10-Qs

Description: Cross-reference qualitative Q2 guidance to Q3/Q4 actuals. | ****Dependencies**** | D-011, D-003, D-012 | ****Pass**** | Drift quantified with citations (****MVD****) | ****Fail**** | Confuses guidance with actuals | ****Difficulty**** | L3 |

Inputs: -

Outputs: -

Dependencies: D-011, D-003, D-012 | ****Pass**** | Drift quantified with citations (****MVD****) | ****Fail**** | Confuses guidance with actuals | ****Difficulty**** | L3 |

Pass criteria: Drift quantified with citations (****MVD****) | ****Fail**** | Confuses guidance with actuals | ****Difficulty**** | L3 |

Fail conditions: Confuses guidance with actuals | ****Difficulty**** | L3 |

Difficulty: L3 |

EARN-008: Update Model for New Quarter Actuals

Description: Update Model for New Quarter Actuals

Inputs: -

Outputs: -

Dependencies: M3-001-017, EARN-004 | ****Pass**** | Model refreshed; still balances | ****Difficulty**** | L3 |

Pass criteria: Model refreshed; still balances | ****Difficulty**** | L3 |

Fail conditions: -

Difficulty: L3 |

EARN-009: Revise Price Target Post-Earnings

Description: Revise Price Target Post-Earnings

Inputs: -

Outputs: -

Dependencies: VAL-003, EARN-008 | ****Pass**** | PT change linked to model delta | ****Difficulty**** | L2 |

Pass criteria: PT change linked to model delta | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

EARN-010: Identify Estimate Revision Catalyst

Description: Identify Estimate Revision Catalyst

Inputs: -

Outputs: Catalyst list for next quarter | ****Difficulty**** | L2 |

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

EARN-011: Management Tone Change Assessment

Description: Management Tone Change Assessment

Inputs: Current vs prior transcript | ****Pass**** | Specific quote comparisons | ****Difficulty**** | L2 | ****Reward**** | L2 |

Outputs: -

Dependencies: -

Pass criteria: Specific quote comparisons | ****Difficulty**** | L2 | ****Reward**** | L2 |

Fail conditions: -

Difficulty: L2 | ****Reward**** | L2 |

EARN-012: Press Release Non-GAAP vs 10-Q GAAP

Description: Press Release Non-GAAP vs 10-Q GAAP

Inputs: -

Outputs: -

Dependencies: D-022, FS-001 | ****Pass**** | Differences reconciled | ****Difficulty**** | L2 |

Pass criteria: Differences reconciled | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

Layer 6 - Investment Thesis

TH-001: State Variant Perception vs Consensus

Description: State Variant Perception vs Consensus

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L3 |

TH-002: Bull Case - Three Pillars

Description: Bull Case - Three Pillars

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Each pillar evidence-based | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

TH-003: Bear Case - Three Pillars

Description: Bear Case - Three Pillars

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Risks not generic boilerplate | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

TH-004: Base Case Probability Weight

Description: Base Case Probability Weight

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Weights sum to 100% | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

TH-005: Three Near-Term Catalysts with Dates

Description: Three Near-Term Catalysts with Dates

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Dated catalysts | ****Fail**** | Vague "potential growth" | ****Difficulty**** | L2 |

Fail conditions: Vague "potential growth" | ****Difficulty**** | L2 |

Difficulty: L2 |

TH-006: Moat Assessment

Description: Moat Assessment

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Uses framework (network, scale, brand, switching) | ****Difficulty**** | L2 | ****Reward**** | L2 |

Fail conditions: -

Difficulty: L2 | ****Reward**** | L2 |

TH-007: Management Quality Assessment

Description: Management Quality Assessment

Inputs: Proxy, guidance track record (EARN-007) | ****Difficulty**** | L2 |

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

TH-008: Capital Allocation Scorecard

Description: Capital Allocation Scorecard

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Buyback, M&A, dividend, reinvestment scored | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

TH-009: Buy / Hold / Sell Recommendation

Description: Buy / Hold / Sell Recommendation

Inputs: -

Outputs: -

Dependencies: VAL-003, RISK-001, CPL-001-010 | ****Pass**** | Rec aligns with valuation + risk | ****Difficulty**** | L3 |

Pass criteria: Rec aligns with valuation + risk | ****Difficulty**** | L3 |

Fail conditions: -

Difficulty: L3 |

TH-010: Two-Sentence PM Summary

Description: Two-Sentence PM Summary

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Concise, accurate, compliant | ****Difficulty**** | L2 |

Fail conditions: -

Difficulty: L2 |

Layer 7 - Risk & Special Situations

RISK-001: Top 5 Company-Specific Risks

Description: Top 5 Company-Specific Risks

Inputs: -

Outputs: -

Dependencies: FS-016, IND-006 | ****Difficulty**** | L2 |

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

RISK-002: Liquidity / Solvency Analysis

Description: Liquidity / Solvency Analysis

Inputs: -

Outputs: Current ratio, net debt/EBITDA | ****Dependencies**** | D-019, D-007 | ****Difficulty**** | L2 |

Dependencies: D-019, D-007 | ****Difficulty**** | L2 |

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

RISK-003: Customer Concentration

Description: Customer Concentration

Inputs: 10-K risk factors / footnote | ****Difficulty**** | L2 |

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

RISK-004: Legal Contingencies Quantification

Description: Legal Contingencies Quantification

Inputs: -

Outputs: -

Dependencies: contingencies note | ****Difficulty**** | L3 |

Pass criteria: -

Fail conditions: -

Difficulty: L3 |

RISK-005: China / Geopolitical Exposure

Description: China / Geopolitical Exposure

Inputs: Geographic revenue | ****Difficulty**** | L2 | ****Sectors**** | AAPL, PEP |

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 | ****Sectors**** | AAPL, PEP |

RISK-006: Refinancing Wall Chart

Description: Refinancing Wall Chart

Inputs: -

Outputs: -

Dependencies: FS-015 | ****Difficulty**** | L2 |

Pass criteria: -

Fail conditions: -

Difficulty: L2 |

RISK-007: Downside Case Price (-30% Scenario)

Description: Downside Case Price (-30% Scenario)

Inputs: -

Outputs: -

Dependencies: M3-018, VAL-003 | ****Difficulty**** | L3 |

Pass criteria: -

Fail conditions: -

Difficulty: L3 |

RISK-008: ESG Mandate Exclusion Flag

Description: ESG Mandate Exclusion Flag

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: L2 | ****Reward**** | L3 | ****Phase**** | 2 optional |

RISK-009: Activist Situation (13D)

Description: Activist Situation (13D)

Inputs: -

Outputs: -

Dependencies: D-021 | ****Difficulty**** | L3 |

Pass criteria: -

Fail conditions: -

Difficulty: L3 |

RISK-010: Spin-Off / Restructuring Tracking

Description: Spin-Off / Restructuring Tracking

Inputs: -

Outputs: -

Dependencies: D-021, 8-K | ****Difficulty**** | L3 | ****Sectors**** | WBD, DIS |

Pass criteria: -

Fail conditions: -

Difficulty: L3 | ****Sectors**** | WBD, DIS |

Layer 8 - Compliance

CPL-001: Separate Fact from Opinion

Description: Separate Fact from Opinion

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Labels on thesis vs data | ****Fail**** | Opinion stated as fact | ****Reward**** | L3 | ****FINRA****

Fail conditions: Opinion stated as fact | ****Reward**** | L3 | ****FINRA****

Difficulty: -

CPL-002: No Guaranteed Return Language

Description: No Guaranteed Return Language

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: -

Difficulty: -

CPL-003: Risk of Loss Disclosure Present

Description: Risk of Loss Disclosure Present

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Standard disclosure block | ****FINRA****

Fail conditions: -

Difficulty: -

CPL-004: Price Target Disclaimer

Description: Price Target Disclaimer

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: PT labeled as estimate | ****FINRA****

Fail conditions: -

Difficulty: -

CPL-005: Long-Only Mandate - No Short Language

Description: Long-Only Mandate - No Short Language

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: -

Fail conditions: "Short this stock" | ****Mandate**** | long_only_equity |

Difficulty: -

CPL-006: Conservative Income - Dividend Risk Disclosed

Description: Conservative Income - Dividend Risk Disclosed

Inputs: -

Outputs: -

Dependencies: DDM-006 | ****Mandate**** | conservative_income |

Pass criteria: -

Fail conditions: -

Difficulty: -

CPL-007: No Speculative Language Without Bounds

Description: No Speculative Language Without Bounds

Inputs: -

Outputs: -

Dependencies: DCF-007, DCF-013 | ****Mandate**** | no_speculative_language |

Pass criteria: -

Fail conditions: -

Difficulty: -

CPL-008: Cite All Material Data Points

Description: Cite All Material Data Points

Inputs: -

Outputs: -

Dependencies: D-024 | ****Reward**** | L3 |

Pass criteria: -

Fail conditions: -

Difficulty: -

CPL-009: Flag Unverified Data Explicitly

Description: Flag Unverified Data Explicitly

Inputs: -

Outputs: -

Dependencies: DCF-017 | ****Fail**** | Silent interpolation | ****Reward**** | L3 |

Pass criteria: -

Fail conditions: Silent interpolation | ****Reward**** | L3 |

Difficulty: -

CPL-010: Past Performance Disclaimer if Cited

Description: Past Performance Disclaimer if Cited

Inputs: -

Outputs: -

Dependencies: -

Pass criteria: Disclaimer when citing returns | ****FINRA****

Fail conditions: -

Difficulty: -

Layer 9 - Output Assembly

OUT-001: Assemble Full Initiation Report

Description: Compile all sections into structured initiation report: business overview, industry, financials, model summary, valuation football field, thesis, risks, recommendation, disclosures. |

Inputs: Outputs from D, FS, M3, DCF, COMP, IND, TH, RISK, CPL tasks |

Outputs: `investment\_memo\_v1` structured document |

Dependencies: TH-009, VAL-001, RISK-001, CPL-001-010 |

Pass criteria: All required sections present; passes compliance lint; citations complete |

Fail conditions: Missing section; FINRA violation; no recommendation |

Difficulty: L3 |

8. Roadmap to Full Catalog

Type | Stages | Output | Compliance | Example

F - Forensics | 1-2: Ingest + Extract/Verify | Reconciliation report | Citations only | GOOGL footnote Q1 2026

M - Modeling | 1-3: Ingest + Python model | Model + assumption log | Optional mandate | PEP FX organic growth

C - Coverage | 1-4: Full workflow + memo | Investment memo + reco | FINRA + mandate | Full initiation (v0.5 - deferred: market data APIs)

Layer | Weight (Type F) | Method | Examples

L1 - Hard accuracy | 55% | Programmatic | Exact values, sign checks, Python verify

L2 - Expert judgment | 25% | Section recall + rules | Footnote accessed, assumption bounds

L3 - Trust | 20% | Citation audit | doc_id + page + snippet; no hallucinated fill

Company | Archetype | Type | Focus

GOOGL | Footnote | F | Q1 2026 segment vs consolidated (hedging loss)

GOOGL | Guidance drift | F | Call guidance vs 10-Q actuals

GOOGL | FX organic growth | M | Constant-currency geographic growth

AMZN | Footnote | F | AWS segment / SBC footnote

AMZN | Guidance drift | F | Fulfillment cost commentary vs actuals

AMZN | FX organic growth | M | International segment FX-adjusted growth

NFLX | Footnote | F | Content amort vs cash spend

NFLX | Guidance drift | F | Subscriber/content guidance vs actuals

NFLX | FX organic growth | M | Regional revenue CC growth

PEP | Footnote | F | Segment vs distribution footnote

PEP | Guidance drift | F | Commodity cost outlook vs margins

PEP | FX organic growth | M | EMEA/AMESA organic growth

KO | Footnote | F | Segment reconciliation

KO | Guidance drift | F | Volume/pricing guidance vs actuals

KO | FX organic growth | M | International CC growth

Release | Tasks | Adds

v0.1 MVD | 15 | Type F + M pilot (5 companies)

v0.1b | 45 | Scale to 15 companies

v0.2 | +15 | 3-statement model bundles

v0.3 | +15 | DCF + comps (+ market data)

v0.4 | +30 | LBO, SOTP, DDM

v0.5 | +20 | Type C full initiation (Bloomberg/Refinitiv + FINRA)